MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5725W[1/]

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTIONS: Pentode, miniature, dual control, RF sharp cutoff

Outline --- 5-1(EIA)
Base --- E7-1
Envelope --- T5-½
Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	k sd	h	h	a	g2	g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rgl	Ik	Ic1
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Meg	mAdc	mAdc
Maximum:	6.9	200	0, -55	155	30, -55	100	0.1	20	1.0
Minimum:	5.7	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	120	-2.0	120	0	0	---	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ic3	Pp	Pg2	TE	Alt
Unit:	mAdc	W	W	°C	ft
Maximum:	0.2	1.65	0.55	165	(see note 1)
Minimum:	---	---	---	---	
TEST CONDITIONS:	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

1/ Formerly tube type 5725

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS		Unit
						Min	Max	
	Quality conformance inspection, part 1							
1256	Electrode current (1) (anode)	See note 2	0.4	II	Ib	2.5	9.0	mAdc
1266	Total grid current	See notes 2 and 3	0.4	II	Ic1	0	-0.1	µAdc
1301	Heater current		0.4	II	If	160	190	mA
1306	Transconductance (1)		0.4	II	Sm (g1-p)	2,500	4,500	µmhos
1336	Heater-cathode leakage		0.4	II	Ihk	---	10	µAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		2.5	I	---	---	---	---
1031	Low-frequency vibration	Rp = 10,000 ohms; 10 G; F = 40 Hz	6.5	Code H	Ep	---	150	mVac
2201	Noise and microphonics	Ef = 6.3 Vdc; Ebb= Ecc2 = 200 Vdc; Ec1 = 0; Eca1 = 175 mVac; Rk = 1,000 ohms; Rp = 0.1 Meg; Rg2 = 0.5 Meg; Cg2 = 2 µF	2.5	I	---	---	---	---
1256	Electrode current (2) (anode)	Ec1 = -3 Vdc; Ec3 = -10 Vdc	2.5	I	Ib	---	200	µAdc
1256	Electrode current (3) (anode)	Ec1 = -3 Vdc; Ec3 = -6 Vdc	2.5	I	Ib	5	---	µAdc
1256	Electrode current (4) (anode)	Ec1 = -8 Vdc	2.5	I	Ib	---	200	µAdc
1256	Electrode current (5) (anode)	Ec1 = -5.5 Vdc	2.5	I	Ib	5	---	µAdc
1256	Electrode current (screen)		2.5	I	Ic2	1.5	5.5	mAdc
1266	Grid emission	Ef = 7.5 V; Ec1 = -10 Vdc; Rg1 = 0.1 Meg (see note 4)	2.5	I	Ic	0	-1.0	µAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS Min	LIMITS Max	Unit
	Quality conformance inspection, part 2 - Continued							
1306	Transconductance (2)	Ef = 5.7 V	2.5	I	$\frac{\Delta Sm\ (g1-p)}{Ef}$	---	15	%
1306	Tranconductance (3)	Ec3 = -3 Vdc	6.5	S3	Sm (g3-p)	350	1,050	µmhos
1306	Transconductance (4)	Ec3 = -5 Vdc	6.5	S3	Sm (g1-p)	700	1,700	µmhos
1331	Direct-interelectrode capacitance	Shield No. 316	6.5	Code E	Cg1p Cin Cout	--- 3.5 2.6	0.02 4.5 3.4	pF pF pF
1121	Base strain	See note 5	---	---	---	---	---	---
2166	Glass strain		2.5	I	---	---	---	---
1041	Shock	450 G; Ehk = +100 Vdc (see note 6)	6.5	See note 7	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 (min), 60 (max)	6.5	See note 7	---	---	---	---
---	Post-shock and vibration-fatigue test end points							
1031	Low-frequency vibration		---	---	Ep	---	300	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	30	µAdc
1306	Transconductance (1)		---	---	Sm(1) (g1-p)	2,200	---	µmhos
1266	Total grid current		---	---	Ic1	0	-0.4	µAdc
1105	Permanence of marking		---	---	---	---	---	---

METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
1506	Heater-cycling life	Ef = 7.5 V; Ehk = +135 Vdc; Ec1 = Ec2 = Ec3 = Eb = 0	---	---	---	---
---	Heater-cycling life-test end point					
1336	Heater-cathode leakage		Ihk	---	20	µAdc
1516	Stability life	Eb = 180 Vdc; Ec1 = 0; Ec2 = 125 Vdc; Ehk = +135 Vdc; Rk = 130 ohms; Rg1 = 0.1 Meg; TA = room	---	---	---	---
---	Stability life-test end point					
1306	Change in transconductance (1) of individual tubes		$\frac{\Delta Sm\ (g1-p)}{t}$	---	10	%

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3 - Continued					
1501	Intermittent life	Stability life-test conditions; TE = +165°C (min) (see notes 8 and 9)	---	---	---	---
---	Intermittent life-test end points (1,000 hours)					
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic1	0	-0.1	µAdc
1301	Heater current		If	160	190	mA
1306	Change in transconductance (1) of individual tubes		$\Delta Sm_{(g1-p)t}$	---	25	%
1306	Transconductance (2)		$\Delta Sm_{(g1-p)Ef}$	---	20	%
1336	Heater-cathode leakage		Ihk	---	10	µAdc
1211	Insulation of electrodes		R	40	---	Meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test shall be performed at the conclusion of the holding period.
3. Approved alternate test. The insertion of a 1.0 megohm resistor in the grid circuit shall change the anode current by not more than 0.5 mAdc.
4. Prior to this test, the tube shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test at specified condition within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ec1	Ec2	Ec3	Eb	Rk	Rg1
V	Vdc	Vdc	Vdc	Vdc	ohms	Meg
7.5	0	125	0	180	130	0.1

5. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (method 1121), except that data covered in "Acceptance rejection criteria" shall be modified as follows:
   a. Accepted if not more than one defective for class "A", "B", or "C" defects, respectively (see method 1121), or if not more than a total of two defectives are found in the sample.
   b. Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.
6. A grid resistor of 0.1 megohm shall be added to each section: however, this resistor shall not be used when a thyratron-type indicator is employed.
7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.

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NOTES: -Continued

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure or more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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